

CS 237: PROBABILITY IN COMPUTING

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Independence for Random Variables

The culprit had big eyes, big ears,
and big teeth. Statistically, the chance
of it being anyone other than the defendant
is only one in 12 million.



REMINDER: INDEPENDENT EVENTS

Definition: Independent Events

Two events A and B are **independent** if

$$\Pr(A \cap B) = \Pr(A) \cdot \Pr(B)$$

- ▶ Sometimes this is called the **product rule**.

INDEPENDENCE: LAST TIME

- ▶ Independence is a very strong assumption
- ▶ Many incorrect arguments in probability are due to incorrect assumptions on the independence of events



CONDITIONAL PROBABILITY: LAST TIME

- ▶ Many incorrect arguments in probability are due to not accounting properly for all the information available
- ▶ Need to make sure we are conditioning on the right events



INDEPENDENT RANDOM VARIABLES

INDEPENDENT RANDOM VARIABLES

- ▶ **Definition:** Random variables X and Y are **independent** if for all $x, y \in \mathbb{R}$, the events $X \leq x$ and $Y \leq y$ are independent.
- ▶ Equivalently: for all $x, y \in \mathbb{R}$, $\mathbb{P}[X \leq x, Y \leq y] = \mathbb{P}[X \leq x]\mathbb{P}[Y \leq y]$

independence of the CDFs at all points

INDEPENDENT RANDOM VARIABLES

- ▶ **Definition:** Random variables X and Y are **independent** if for all $x, y \in \mathbb{R}$, the events $X \leq x$ and $Y \leq y$ are independent.
- ▶ Equivalently: for all $x, y \in \mathbb{R}$, $\mathbb{P}[X \leq x, Y \leq y] = \mathbb{P}[X \leq x]\mathbb{P}[Y \leq y]$
- ▶ This definition holds for both **discrete and continuous** random variables. *(because both discrete & continuous r.v. have CDFs!)*
- ▶ For discrete random variables, *it is equivalent to the following* for all $x, y \in \mathbb{R}$:
$$\mathbb{P}[X = x, Y = y] = \mathbb{P}[X = x]\mathbb{P}[Y = y]$$

ASK THE AUDIENCE

- ▶ **Experiment:** toss a fair coin 3 times (independently)
- ▶ C = number of heads
- ▶ $M = 1$ if all the tosses match and $M = 0$ otherwise

Are C and M independent?

(A) Yes

(B) No

ASK THE AUDIENCE

- ▶ **Experiment:** toss a fair coin 3 times (independently)
- ▶ C = number of heads
- ▶ $M = 1$ if all the tosses match and $M = 0$ otherwise

if $C=3$, then $M=1$!

$$\Rightarrow \mathbb{P}[C=3, M=0] = 0$$

$$\text{but, } \mathbb{P}[C=3] - \mathbb{P}[M=0] > 0 \neq 0$$

ASK THE AUDIENCE

- ▶ **Experiment:** toss a fair coin 3 times (independently)
- ▶ $H_1 = 1$ if the first toss is H, and $H_1 = 0$ otherwise
- ▶ $M = 1$ if all the tosses match and $M = 0$ otherwise

Are H_1 and M independent?

(A) Yes

(B) No

ASK THE AUDIENCE

- ▶ **Experiment:** toss a fair coin 3 times (independently)
- ▶ $H_1 = 1$ if the first toss is H, and $H_1 = 0$ otherwise
- ▶ $M = 1$ if all the tosses match and $M = 0$ otherwise

$$P[H_1 = 1] = P[H_1 = 0] = \frac{1}{2} \quad P[M = 1] = P[\{HHH, TTT\}]$$

$$= \frac{2}{8} = \frac{1}{4}$$

$$P[M = 0] = \frac{3}{4}$$

$$P[H_1 = 1, M = 1] = P[\{HHH\}] = \frac{1}{8} = \frac{1}{2} \cdot \frac{1}{4} \quad \checkmark$$

$$P[H_1 = 1, M = 0] = P[\{HHT, HTH, HTT\}] = \frac{3}{8} = \frac{1}{2} \cdot \frac{3}{4} \quad \checkmark$$

Similar computations for $P[H_1 = 0, M = 1]$ and $P[H_1 = 0, M = 0]$ 

MUTUAL INDEPENDENCE

- ▶ The notion of mutual independence similarly carries over to random variables
- ▶ **Definition** (mutual independence for n random variables): Random variables $X_1, X_2, \dots, X_n$ are mutually independent if for all values $x_1, x_2, \dots, x_n \in \mathbb{R}$, the events

$$[X_1 \leq x_1], [X_2 \leq x_2], \dots, [X_n \leq x_n]$$

are mutually independent

MUTUAL INDEPENDENCE

- ▶ If $X_1, X_2, \dots, X_n$ are **discrete** random variables, the definition is equivalent to asking for all values $x_1, x_2, \dots, x_n \in \mathbb{R}$ we have that the events

$$[X_1 = x_1], [X_2 = x_2], \dots, [X_n = x_n]$$

are mutually independent

- ▶ Notice it is sufficient to only check x_i in the range of X_i

ASK THE AUDIENCE

- ▶ Let X and Y be **independent** random variables, each taking on the values -1 and 1 with probability $1/2$
- ▶ Let $Z = X \cdot Y$

Are X, Y, Z **pairwise** independent?

(A) Yes

(B) No

ASK THE AUDIENCE

- ▶ Let X and Y be **independent** random variables, each taking on the values -1 and 1 with probability $1/2$
- ▶ Let $Z = X \cdot Y$

- X, Y are pairwise indep by definition

- X, Z : if I know that $X=1$, Z is still equally likely to be -1 or 1 .

$$\mathbb{P}[Z=1, X=1] = \mathbb{P}[X=Y=1] = 1/4 = \mathbb{P}[Z=1] \mathbb{P}[X=1] \quad \checkmark$$

Other cases are similar.

ASK THE AUDIENCE

- ▶ Let X and Y be **independent** random variables, each taking on the values -1 and 1 with probability $1/2$
- ▶ Let $Z = X \cdot Y$

Are X, Y, Z **mutually** independent?

(A) Yes

(B) No

ASK THE AUDIENCE

- ▶ Let X and Y be **independent** random variables, each taking on the values -1 and 1 with probability $1/2$
- ▶ Let $Z = X \cdot Y$

if $X=Y=1$, we know $Z=1$! And $P(Z=1) \neq 1$.

So they are not independent.

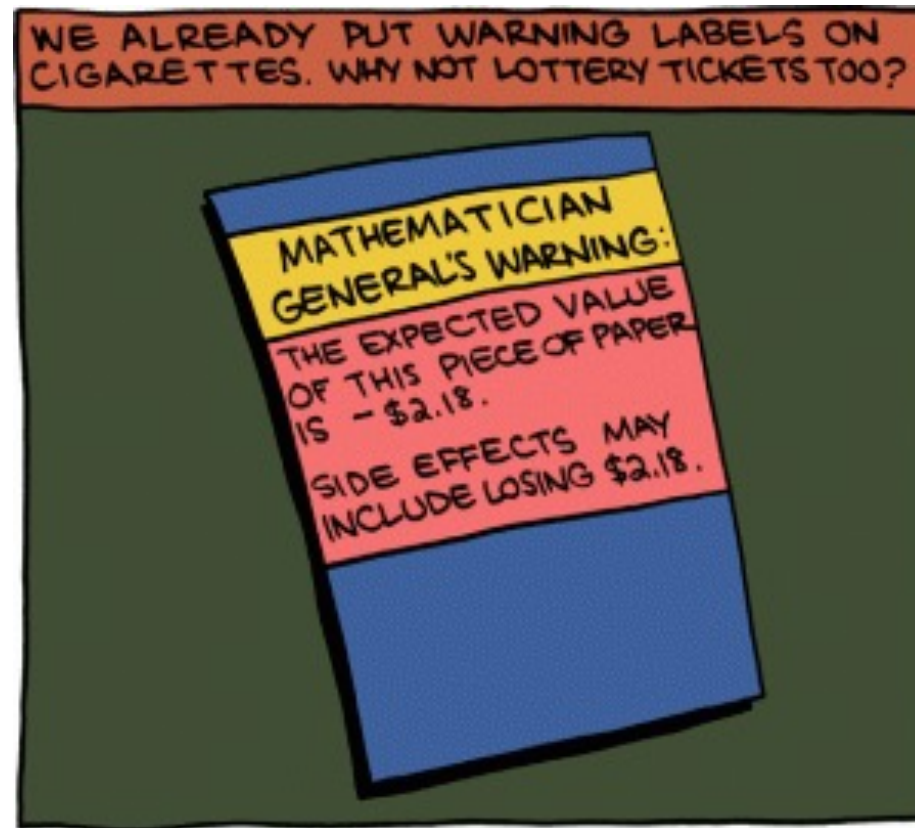
$$P(Z=-1, X=1, Y=1) = 0 \neq P(Z=-1)P(X=1)P(Y=1)$$

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Introduction to Expected Value



<https://www.smbc-comics.com/comic/2011-12-04>

SUMMARY STATISTICS

- ▶ Motivating example: household income in the US
- ▶ **Approach 1:** do a census and obtain all the incomes
 - ▶ **Pros:** provides the most information, can perform any analysis
 - ▶ **Cons:** difficult/expensive to obtain, we may only be interested in a summary to begin with
- ▶ **Approach 2:** obtain a suitable summary of the incomes

SUMMARY STATISTICS

- ▶ Motivating example: household income in the US
- ▶ We want to provide some information about the incomes
- ▶ **Examples:** minimum, maximum, median, average, ...
- ▶ Each of them is useful, a combination of them even better
- ▶ Some of them are difficult to estimate (e.g. min, max)
- ▶ The *average* can be estimated very well using a small number of samples

SUMMARY STATISTICS

- ▶ Let X be a discrete random variable
 - ▶ Example: X = income of a random household
- ▶ One of the most important summary statistics for X is its “average” value (the expectation or mean of X)
- ▶ **Plan for today:** definition and properties of expectation
- ▶ **Later:** how to estimate expectation by sampling

SUMMARY STATISTICS

- ▶ Let X be a discrete random variable
 - ▶ Example: X = income of a random household
- ▶ One of the most important summary statistics for X is its “average” value (the expectation or mean of X)
- ▶ **Plan for today:** definition and properties of expectation
- ▶ **Later:** how to estimate expectation by sampling

Many applications: polling for elections, census data, product testing, etc.

EXPECTATION

- ▶ **Definition:** Let X be a discrete random variable. The expected value (or expectation or mean) of X is

$$\mathbb{E}(X) = \sum_{\omega \in \Omega} X(\omega) \cdot \mathbb{P}[\{\omega\}]$$

- ▶ Expectation is a weighted average, where each outcome is weighted by its probability

EXPECTATION

- **Definition:** Let X be a discrete random variable. The expected value (or expectation or mean) of X is

$$\mathbb{E}(X) = \sum_{\omega \in \Omega} X(\omega) \cdot \mathbb{P}[\{\omega\}]$$

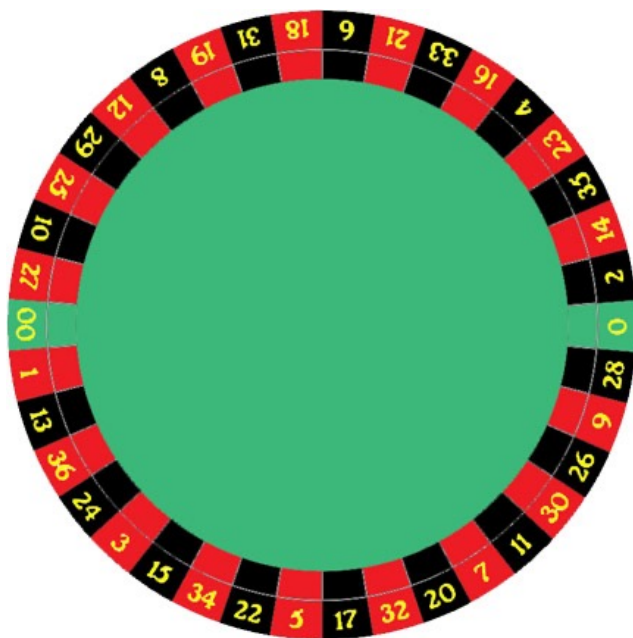
recall
 $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

X = number obtained when rolling a die

$$\mathbb{E}(X) = \sum_{i=1}^6 i \cdot \frac{1}{6} = \frac{1}{6} \cdot \sum_{i=1}^6 i = \frac{1}{6} \cdot \frac{6 \cdot 7}{2} = \boxed{\frac{7}{2}}$$

EXAMPLE: ROULETTE

- 38 slots: 18 black, 18 red, 2 green.
- If we bet \$1 on red, we get \$2 back if red comes up. What's the expected return of this game?



$X = \text{change in \$}$

$$E[X] = \underbrace{\frac{18}{38}(-1)}_{P[\text{black}]} + \underbrace{\frac{18}{38}(1)}_{P[\text{red}]} + \underbrace{\frac{2}{38}(-1)}_{P[\text{green}]}$$

$$= \boxed{-\frac{2}{38}}$$

EXPECTATION

- By rearranging the terms in the definition, we obtain the following equivalent way to write the expectation:

$$\mathbb{E}(X) = \sum_{a \in \text{range}(X)} a \cdot \mathbb{P}[X = a]$$

just change the order of summation

$\sum_{a=1}^{\infty}$

1	1	1
0	0	0
2	2	2
0	0	0

$a=2$

Proof:

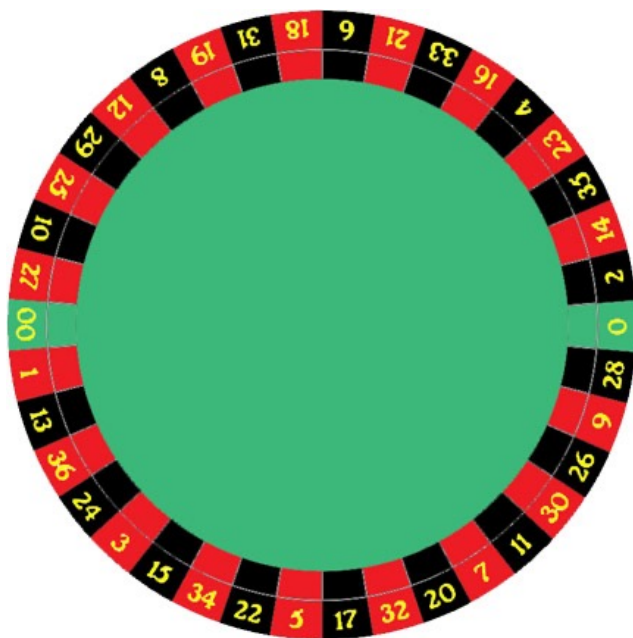
$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) \cdot \mathbb{P}[\{\omega\}] = \sum_{a \in \text{range}(X)} \sum_{\substack{\omega \in \Omega: \\ X(\omega) = a}} X(\omega) \cdot \mathbb{P}[\{\omega\}]$$

$(\mathbb{P}[X=a])!$

$$= \sum_{a \in \text{range}(X)} a \cdot \left[\sum_{\substack{\omega \in \Omega \\ X(\omega) = a}} \mathbb{P}[\{\omega\}] \right] = \sum_{a \in \text{range}(X)} a \cdot \mathbb{P}[X=a]$$

EXAMPLE: ROULETTE

- ▶ If we bet \$1 on **red**, we get \$2 back if **red** comes up. What's the expected return of this game?
- ▶ Let X be the amount we receive minus our bet



$$\mathbb{E}(X) = \sum_{a \in \text{range}(X)} a \cdot \mathbb{P}[X = a]$$

$$\mathbb{E}(X) = \underbrace{1}_{\text{and on red}} \cdot \frac{18}{38} + \underbrace{(-1)}_{\text{otherwise}} \cdot \frac{20}{38} = -\frac{2}{38} \quad \checkmark$$

EXAMPLE: COIN TOSS

- ▶ Toss a coin that comes up Heads with probability p
- ▶ Let $X = \begin{cases} 1 & \text{if Heads} \\ 0 & \text{if Tails} \end{cases}$

Classic example you will see over and over!

$$E[X] = 1 \cdot p + 0 \cdot (1-p) = \underline{p}$$

EXPECTATION

- ▶ The expectation of a continuous random variable is defined similarly to the discrete case, only that we replace the sum by an integral
- ▶ Let X be a random variable with PMF f (discrete) or PDF f (continuous)

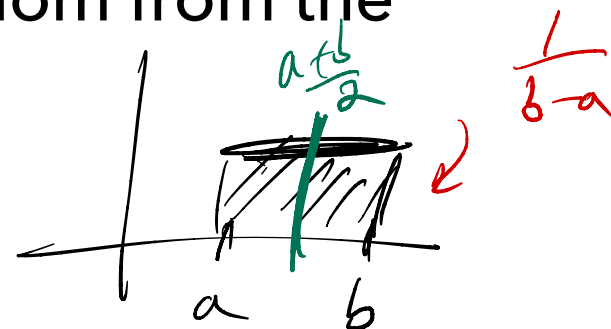
discrete:
$$\mathbb{E}(X) = \sum_{x \in \mathbb{R}} x \cdot f(x)$$

continuous:
$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

EXAMPLE: UNIFORM RANDOM VARIABLE

- Let X be a value chosen uniformly at random from the interval $[a, b]$

- The PDF is $f(x) = \frac{1}{b-a}$ for $x \in [a, b]$



$$\begin{aligned}
 \mathbb{E}[X] &= \int_{-\infty}^{\infty} x \cdot f(x) dx = \int_a^b x \cdot \frac{1}{b-a} dx \\
 &= \frac{1}{b-a} \left. \frac{x^2}{2} \right|_a^b = \frac{1}{b-a} \left(\frac{b^2}{2} - \frac{a^2}{2} \right) \\
 &= \left(\frac{1}{b-a} \right) \left(\frac{1}{2} \right) (b-a)(b+a) = \boxed{\frac{b+a}{2}} \checkmark
 \end{aligned}$$

ASK THE AUDIENCE: DARTS

- ▶ Suppose we throw a dart at a circular board of radius r
- ▶ Let D be the distance from the center of the board to the hitting point of the dart
- ▶ The PDF is $f(x) = \frac{2x}{r^2}$ for $x \in [0, r]$
- ▶ What is $\mathbb{E}[D]$?

(A) 1

(B) $\frac{r}{2}$ (C) $\frac{r}{3}$ (D) $\frac{2r}{3}$

ASK THE AUDIENCE: DARTS

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- ▶ Let D be the distance from the center of the board to the hitting point of the dart
- ▶ The PDF is $f(x) = \frac{2x}{r^2}$ for $x \in [0, r]$
- ▶ What is $\mathbb{E}[D]$?

$$\mathbb{E}[D] = \int_0^r x \cdot \frac{2x}{r^2} dx = \frac{2}{r^2} \int_0^r x^2 dx = \frac{2}{r^2} \frac{r^3}{3} = \boxed{\frac{2r}{3}}$$

NEXT TIME: LINEARITY OF EXPECTATION

- ▶ **Theorem:** For **any** random variables X and Y

$$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$$

- ▶ The theorem holds even if X and Y are dependent!
- ▶ We will discuss this more last time.

